PHARMACOLOGY · MCQ PRACTICE

Emetic and Antiemetic Drugs — MCQ Practice

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GIT BLOCK

Pharmacology GIT Block · MEDucation by Mattin Majid

HOW TO USE THIS SET

20 questions, 5 options each (A–E). Answer on paper or in your head, then check every answer against the key on the final page.

4 Easy · 10 Medium · 6 Hard · suggested time ≈ 25 minutes.

Q Questions

1. Which receptor do 5-HT3 antagonists (e.g. ondansetron) block to control vomiting? EASY

- **A.** D2 dopamine receptors
- **B.** H1 histamine receptors
- **C.** 5-HT3 serotonin receptors
- **D.** M muscarinic receptors
- **E.** NK1 neurokinin receptors

2. Which drug class is the classic choice for prophylaxis of motion sickness? EASY

- **A.** 5-HT3 antagonists
- **B.** NK1 antagonists
- **C.** Muscarinic antagonists (e.g. hyoscine)
- **D.** Glucocorticoids
- **E.** Benzodiazepines

3. What is the primary mechanism of metoclopramide's antiemetic action? EASY

- **A.** 5-HT3 receptor blockade
- **B.** D2 dopamine receptor antagonism centrally and peripherally
- **C.** H1 receptor blockade
- **D.** NK1 receptor blockade
- **E.** Direct gastric mucosal irritation

4. Which class of antiemetics is particularly effective for the DELAYED phase of chemotherapy-induced nausea and vomiting? EASY

- **A.** H1 antagonists
- **B.** NK1 antagonists (e.g. aprepitant)
- **C.** Muscarinic antagonists
- **D.** Emetics
- **E.** 5-HT3 antagonists alone

5. A Parkinson's disease patient on levodopa develops nausea and needs an antiemetic. Metoclopramide would be a poor choice because it would likely: MEDIUM

- **A.** Have no antiemetic effect in Parkinson's patients
- **B.** Cross into the CNS and worsen parkinsonian symptoms via central D2 blockade
- **C.** Cause severe QT prolongation specifically in Parkinson's patients
- **D.** Interact dangerously with levodopa via CYP3A4
- **E.** Cause hypertension as its main risk

6. A patient receiving high-dose cisplatin chemotherapy is placed on a dexamethasone + ondansetron regimen for the first 24 hours. What is the expected clinical benefit of adding dexamethasone to a 5-HT3 antagonist? MEDIUM

A. Dexamethasone has no additive antiemetic benefit
B. The combination significantly boosts antiemetic response rate compared with either drug alone
C. Dexamethasone is added only to treat 5-HT3-antagonist-induced rash
D. Dexamethasone replaces the need for a 5-HT3 antagonist entirely
E. Dexamethasone is added solely to prevent QT prolongation

7. A young adult started on metoclopramide develops spasmodic torticollis and motor restlessness. What is the underlying mechanism of this adverse effect? MEDIUM

A. 5-HT3 blockade in the gut
B. Central D2 receptor blockade causing extrapyramidal effects, more common in children/young adults
C. H1 receptor blockade causing sedation
D. NK1 receptor blockade
E. Direct muscle toxicity unrelated to dopamine

8. A patient on aprepitant for CINV is also started on clarithromycin for a sinus infection. What is the primary pharmacological concern? MEDIUM

A. Clarithromycin has no interaction with aprepitant
B. Clarithromycin is a strong CYP3A4 inhibitor, which could increase aprepitant levels since aprepitant is CYP3A4-metabolised
C. Clarithromycin reduces aprepitant absorption via chelation
D. Aprepitant inactivates clarithromycin
E. This combination is only a concern in renal impairment

9. A cancer patient with hepatic impairment (but normal renal function) needs dose consideration for ondansetron. What is the correct approach based on its pharmacokinetics? MEDIUM

A. No dose adjustment needed in either hepatic or renal impairment
B. Dose reduction is needed for hepatic impairment, not renal impairment
C. Dose reduction is needed for renal impairment, not hepatic impairment
D. Dose reduction is needed for both hepatic and renal impairment equally
E. Ondansetron is contraindicated in any hepatic impairment

10. A pregnant patient with severe morning sickness needs an antiemetic. Which drug is specifically noted as a good choice for this indication? MEDIUM

A. Aprepitant
B. Promethazine
C. Nabilone
D. Ivermectin (unrelated)
E. Chlorpromazine only

11. A patient taking nabilone for refractory chemo-induced nausea is later given naloxone for an unrelated opioid overdose in the ED. What happens to their nausea control, and what does this reveal about nabilone's mechanism? MEDIUM

A. No change — the two drugs act on unrelated pathways
B. Nabilone's antiemetic effect is antagonised, suggesting opioid receptors contribute to its mechanism despite being a cannabinoid
C. Naloxone potentiates nabilone's effect
D. Naloxone converts nabilone into an emetic agent
E. This combination causes serotonin syndrome

12. A patient with severe kerosene ingestion is mistakenly considered for ipecacuanha to induce vomiting. What is the specific danger of this approach? MEDIUM

A. Ipecacuanha has no emetic effect on hydrocarbons

B. Vomiting hydrocarbons risks aspiration and chemical pneumonitis, so induced emesis is contraindicated in this poisoning

C. Ipecacuanha would neutralize the kerosene chemically

D. Ipecacuanha is only effective in children

E. This combination causes fatal serotonin syndrome

13. A patient is prescribed a phenothiazine (e.g. prochlorperazine) for chemotherapy-induced nausea. Beyond D2 blockade, which additional receptor blockades contribute to its antiemetic and side-effect profile? MEDIUM

A. Only 5-HT3 blockade

B. Histamine and muscarinic receptor blockade, in addition to D2 blockade in the CTZ

C. Only NK1 blockade

D. Beta-adrenergic blockade only

E. GABA receptor blockade only

14. A hospitalized patient develops diarrhea after being started on high-dose metoclopramide plus dexamethasone for CINV. Which added drug is used specifically to counter this metoclopramide-related side effect? MEDIUM

A. An NK1 antagonist

B. A benzodiazepine

C. An antihistamine (e.g. diphenhydramine)

D. A 5-HT3 antagonist

E. A muscarinic antagonist

15. A patient develops nausea from a cause acting directly and exclusively on the CTZ (not the vestibular system or GI irritation). Which of the following classes would be LEAST effective, based on the lecture's described spectrum of activity? HARD

A. 5-HT3 antagonists

B. D2 antagonists

C. NK1 antagonists

D. H1 antagonists — effective against many causes including motion sickness and gastric irritants, but NOT effective against CTZ-direct causes

E. Cannabinoids

16. A clinician is designing a highly emetogenic chemotherapy regimen's antiemetic prophylaxis and wants maximal coverage of both acute and delayed CINV phases. Based on the lecture's described combination strategy, which regimen best achieves this? HARD

A. 5-HT3 antagonist alone

B. Dexamethasone + 5-HT3 antagonist + NK1 antagonist, since adding an NK1 antagonist to a 5-HT3 antagonist + dexamethasone base is specifically useful for delayed CINV in highly emetogenic regimens

C. H1 antagonist + muscarinic antagonist only

D. Benzodiazepine monotherapy

E. Emetic drug followed by an antiemetic

17. A patient with a QT-prolonging cardiac condition needs a 5-HT3 antagonist for postoperative nausea. Among the class, which agent carries a comparatively higher noted risk of QT prolongation, making it a less ideal choice here? HARD

A. Ondansetron equally with all others

B. Dolasetron, noted to have relatively more QT-interval prolongation risk within the class

C. Granisetron exclusively

D. Palonosetron exclusively

E. None of the 5-HT3 antagonists have any QT effect

18. A patient with Parkinson's disease and cytotoxic-induced vomiting is being considered for domperidone, but the physician is concerned about a specific systemic risk at higher doses in older patients. What is this risk? HARD

A. Extrapyramidal symptoms, same as metoclopramide

B. A small increased risk of serious cardiac adverse effects, especially at higher doses and in older patients — leading to restricted use

C. Severe hepatotoxicity requiring liver transplant

D. Irreversible tardive dyskinesia

E. Fatal serotonin syndrome

19. A pharmacology exam question asks students to explain why aprepitant has a complex drug interaction profile beyond simply being a CYP3A4 substrate. What is the additional mechanistic complexity described in the lecture? HARD

A. Aprepitant has no other interaction mechanism beyond being a substrate

B. Aprepitant itself induces CYP3A4/2C9 and dose-dependently inhibits CYP3A4, meaning it is both affected by and affects the same enzyme system, creating bidirectional interaction risk

C. Aprepitant only affects renal transporters, not CYP enzymes

D. Aprepitant's interactions are limited to protein-binding displacement

E. Aprepitant's only interaction risk is with grapefruit juice

20. A researcher compares the four main neurotransmitter/receptor pairs driving emesis (5-HT/5-HT3, ACh/M, dopamine/D2, histamine/H1) and notes that combination antiemetic regimens targeting multiple pathways tend to outperform single-agent therapy for highly emetogenic chemotherapy. What is the best physiological rationale for this observation, based on the lecture's content? HARD

A. All four pathways are functionally identical, so combining them has no rationale, and the observation must be coincidental

B. Multiple parallel/converging pathways drive the vomiting reflex (as shown by the four separate transmitter-receptor pairs each linked to different antiemetic classes), so blocking only one leaves the others free to trigger emesis, explaining why combination regimens (e.g. dexamethasone + 5-HT3 antagonist ± NK1 antagonist) outperform monotherapy

C. Combination therapy works purely by increasing total drug dose, unrelated to receptor diversity

D. Single-agent 5-HT3 blockade is always sufficient and combinations offer no benefit

E. The four pathways are sequential, not parallel, so blocking any one would fully block the reflex

✓ Answer key

Q	Answer	Difficulty	Why
1	C	EASY	5-HT3 antagonists block serotonin receptors, mainly at the CTZ.
2	C	EASY	Hyoscine (scopolamine), a muscarinic antagonist, is used for motion sickness prophylaxis/treatment.
3	B	EASY	Metoclopramide is a D2 antagonist acting on the CTZ and peripherally on the GI tract.
4	B	EASY	NK1 antagonists like aprepitant specifically control the delayed phase of cytotoxic-drug emesis.
5	B	MEDIUM	Metoclopramide crosses the blood-brain barrier and blocks central D2 receptors, which can worsen Parkinsonian symptoms — domperidone is preferred because it does not readily cross the BBB.
6	B	MEDIUM	Corticosteroids boost antiemetic activity when combined with a 5-HT3 antagonist; dexamethasone + ondansetron shows a notably high response rate (91% in the first 24h post-cisplatin).
7	B	MEDIUM	Metoclopramide's central D2 blockade causes extrapyramidal movement disorders, which are more common in children and young adults.
8	B	MEDIUM	Aprepitant/rolapitant are metabolised by CYP3A4; strong CYP3A4 inhibitors like clarithromycin should be used cautiously due to interaction risk.
9	B	MEDIUM	5-HT3 antagonists are excreted by kidney and liver but specifically need dose reduction in liver impairment, not renal impairment.
10	B	MEDIUM	Promethazine is specifically noted for severe pregnancy morning sickness (and was also NASA's choice for space motion sickness).
11	B	MEDIUM	Nabilone's antiemetic effect is antagonised by naloxone, implying that opioid receptors play a role in its mechanism despite being classified as a cannabinoid.
12	B	MEDIUM	

			Ipecacuanha is contraindicated in kerosene/corrosive poisoning because inducing vomiting risks aspiration of the hydrocarbon into the lungs.
13	**B**	MEDIUM	Phenothiazines are mainly D2 antagonists in the CTZ but also block histamine and muscarinic receptors, contributing to both efficacy and side effects (sedation, dry mouth, etc.).
14	**C**	MEDIUM	Antihistamines like diphenhydramine are added to corticosteroids to counter metoclopramide-induced diarrhea, or to high-dose metoclopramide to reduce extrapyramidal reactions.
15	**D**	HARD	H1 antagonists are specifically noted as effective for motion sickness and gastric irritants but not for CTZ-direct causes, unlike 5-HT3, D2, NK1 antagonists, and cannabinoids which act more centrally at/near the CTZ.
16	**B**	HARD	The lecture specifies that adding an NK1 antagonist to a 5-HT3 antagonist + dexamethasone combination is especially useful for highly emetogenic regimens, particularly for delayed CINV — covering both acute (5-HT3/steroid) and delayed (NK1) phases.
17	**B**	HARD	Among the 5-HT3 antagonists, QT-interval prolongation is noted as occurring "more with dolasetron," making it a comparatively less ideal choice in a patient with existing QT concerns.
18	**B**	HARD	Despite sparing the CNS, domperidone carries a small increased risk of serious cardiac adverse effects (especially at higher doses and in older patients), which has led to restricted use.
19	**B**	HARD	Aprepitant itself induces CYP3A4/2C9 and dose-dependently inhibits CYP3A4 — meaning it is metabolised by CYP3A4 (substrate) while also modulating the same enzyme, creating a complex bidirectional interaction profile.
20	**B**	HARD	Since four distinct transmitter/receptor pairs (5-HT/5-HT3, ACh/M, dopamine/D2, histamine/H1) each drive the emetic reflex and each corresponds to a distinct drug class, blocking only one still leaves other convergent pathways active — explaining why multi-pathway combination regimens achieve higher response rates than single-agent therapy, as shown by the documented combination response-rate data.